REFLEXIVITY OF A COHERENT MODULE ON AN INTEGRAL NORMAL SCHEME IS TORSION FREENESS PLUS S2, EQUIVALENTLY EXTENDABILITY FROM A BIG OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 2254-2270 and proof span 2272-2301 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof obtains the two directions of (3) with the same cited criterion for normal local rings, applied once for the depth statement outside the big open and once at the end; both citations are preserved and handled as external navigation pointers.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The affine criterion for reflexivity of a coherent module, cited from the same chapter, which reduces reflexivity to the corresponding statement over local rings. Cited at proof lines 2272-2273. Reduces the equivalence of (1) and (2) to the algebraic statement over normal local rings.
- The extension lemma for coherent modules with property S2 across a closed subset of codimension at least two, cited from the same chapter. Cited at proof lines 2280-2281 and again at 2300-2301. Supplies both directions linking S2 (equivalently sufficient depth outside a big open) with extendability of the restriction from the big open.
- Local rings of points outside a big open in an integral normal locally Noetherian scheme have depth at least two, and rings of dimension at most one there are fields or discrete valuation rings. Cited at proof lines 2276-2278, 2292-2295 and 2296-2299. Gives the depth and dimension statements that identify the complement of the maximal locally free open as being of codimension at least two.

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- A finite locally free module is reflexive and, over a discrete valuation ring or a field, a torsion free module is free. Used at proof lines 2279-2280 and 2296-2299. Provides the easy inclusion (3) implies (1) and the freeness of the stalks at points of local dimension at most one.

Source reference. Original lemma 0AY0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13773–13773. Same-chapter reference used in the retained author text at source line(s) 2273. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0EBJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17924–17924. Same-chapter reference used in the retained author text at source line(s) 2280, 2300, 7349, 7397, 7446, 7506, 7534. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma modules-lemma-finite-presentation-stalk-free: sources/stacks-divisors/pinned-original-source.tex, source lines 2288–2288. Same-chapter reference used in the retained author text at source line(s) 2288. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma more-algebra-lemma-dedekind-torsion-free-flat: sources/stacks-divisors/pinned-original-source.tex, source lines 1822–1822. Same-chapter reference used in the retained author text at source line(s) 1822, 2293, 2361. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma more-algebra-lemma-reflexive-over-normal: sources/stacks-divisors/pinned-original-source.tex, source lines 2275–2275. Same-chapter reference used in the retained author text at source line(s) 2275, 2364. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma properties-lemma-codimension-local-ring: sources/stacks-divisors/pinned-original-source.tex, source lines 2295–2295. Same-chapter reference used in the retained author text at source line(s) 2295, 3518, 6583. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma properties-lemma-criterion-normal: sources/stacks-divisors/pinned-original-source.tex, source lines 2277–2277. Same-chapter reference used in the retained author text at source line(s) 2277, 2291, 2297, 3050, 7350, 7398, 7412, 7507, 7535. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be an integral locally Noetherian normal scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. The following are equivalent*

(1) *$\mathcal{F}$ is reflexive,*
(2) *$\mathcal{F}$ is torsion free and has property (S_2), and*
(3) *there exists an open subscheme $j : U \to X$ such that*
   (a) *every irreducible component of $X \setminus U$ has codimension ≥ 2 in X,*
   (b) *$j^*\mathcal{F}$ is finite locally free, and*
   (c) *$\mathcal{F} = j_*j^*\mathcal{F}$.*

Proof. Using Lemma 0AY0 the equivalence of (1) and (2) follows from More on Algebra, Lemma more-algebra-lemma-reflexive-over-normal. Let $U \subset X$ be as in (3). By Properties, Lemma properties-lemma-criterion-normal we see that $\text{depth}(\mathcal{O}_{X,x}) \geq 2$ for $x \notin U$. Since a finite locally free module is reflexive, we conclude (3) implies (1) by Lemma 0EBJ.

Assume (1). Let $U \subset X$ be the maximal open subscheme such that $j^*\mathcal{F} = \mathcal{F}|_U$ is finite locally free. So (3)(b) holds. Let $x \in X$ be a point. If $\mathcal{F}_x$ is a free $\mathcal{O}_{X,x}$-module, then $x \in U$, see Modules, Lemma modules-lemma-finite-presentation-stalk-free. If $\dim(\mathcal{O}_{X,x}) \leq 1$, then $\mathcal{O}_{X,x}$ is either a field or a discrete valuation ring (Properties, Lemma properties-lemma-criterion-normal) and hence $\mathcal{F}_x$ is free (More on Algebra, Lemma more-algebra-lemma-dedekind-torsion-free-flat). Thus $x \notin U \Rightarrow \dim(\mathcal{O}_{X,x}) \geq 2$. Then Properties, Lemma properties-lemma-codimension-local-ring shows (3)(a) holds. By the already used Properties, Lemma properties-lemma-criterion-normal we also see that $\text{depth}(\mathcal{O}_{X,x}) \geq 2$ for $x \notin U$ and hence (3)(c) follows from Lemma 0EBJ. □